
Discovering Cooperative Pipelines: Autoresearch for Sequential Social Dilemmas

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Abstract

1 We study *two-level autoresearch* for cooperation: an outer-loop AI agent au-
2 tonomously redesigns the inner-loop pipeline of an LLM policy-synthesis sys-
3 tem for multi-agent Sequential Social Dilemmas (SSDs). A *researcher agent* $\mathcal{R}$
4 (run as a coding agent) reads the inner-loop source code, edits system prompts,
5 feedback functions, helper libraries, and iteration logic, runs evaluations, and
6 decides what to keep, following the *autoresearch* paradigm. Across two games
7 (Cleanup and Gathering), two policy-synthesizer LLMs, and two welfare objec-
8 tives (utilitarian efficiency and Rawlsian maximin), the researcher reliably exceeds
9 hand-designed baselines, sharply tightens run-to-run variance, and outperforms
10 prompt-only optimization. The discovered pipelines are objective-dependent: un-
11 der maximin the researcher independently rediscovers *duty rotation* as a fairness
12 mechanism, supporting an *information-design* reading in which the researcher
13 chooses what to reveal to the boundedly rational synthesizer. Code at <https://github.com/anon590/llm-policies-social-dilemmas>.
14

15 1 Introduction

16 Sequential Social Dilemmas (SSDs) [1] are the multi-agent analogue of the prisoner’s dilemma in
17 temporally rich Markov games: individually rational play leads to collectively suboptimal outcomes
18 through pollution, over-harvesting, or open conflict. Standard multi-agent reinforcement learning
19 (MARL) struggles in this regime due to credit assignment, non-stationarity, and large joint action
20 spaces [2]. A complementary approach, recently introduced by Gallego [3], replaces parameter-
21 space optimization with *algorithm-space synthesis*: a frozen LLM writes a Python policy function,
22 evaluates it in self-play, and iteratively refines it from performance feedback. A single generation
23 step can produce coordination logic (territory partitioning, role assignment, conditional cooperation)
24 at a sample efficiency several orders of magnitude beyond what gradient-based MARL achieves on
25 the same environments.

26 This shifts where the design problem lives, rather than removing it. The inner-loop pipeline that
27 drives the synthesizer has many free parameters: which system prompt, which feedback variables,
28 which helper functions, how many refinement steps. Each materially affects the resulting policies,
29 and prior work tuned them by hand. A natural question follows: *can an AI agent design the pipeline?*

30 We answer affirmatively with a two-level autoresearch framework. An *outer-loop researcher agent*
31 (Claude Opus 4.6, run as a coding agent) edits the source files of an *inner-loop policy synthesizer*
32 (another LLM), runs evaluations on held-out seeds, and keeps modifications that improve a fixed
33 welfare objective Φ . The outer agent operates on an ordinary git repository (reading code, writing
34 diffs, running shell commands, etc) without task-specific scaffolding, mirroring the autoresearch
35 paradigm of Karpathy [4] for single-GPU LLM pretraining. Although the inner-loop SSDs are
36 gridworld benchmarks rather than physical systems, the outer-loop discovery process itself runs

under conditions a deployed discovery agent faces: noisy multi-seed evaluations, stochastic code generation, an LLM-evaluation budget that bounds how often Φ can be queried, and a heterogeneous code repository the agent must navigate end-to-end on its own.

Our contributions are: i) a general two-level framework that delegates the design of an LLM synthesis pipeline to a coding agent operating on a real software repository (Section 3); ii) the first instantiation of the autoresearch paradigm in a multi-agent decision-making domain, with experiments across two SSDs (Cleanup and Gathering), two policy LLMs, and two welfare objectives (utilitarian efficiency U and Rawlsian maximin $\min_i R_i$) (Section 4); and iii) a mechanism-design interpretation supported by the qualitatively different pipelines the agent produces under different welfare objectives, including the autonomous discovery of *time-based duty rotation* as a fairness mechanism, never found under efficiency optimization.

2 Background

We build on the iterative LLM policy synthesis framework of Gallego [3], which serves as the (frozen) inner loop of our two-level system. This section recalls the SSD formalism, the social outcome metrics, and the base synthesis loop.

2.1 Sequential Social Dilemmas

A Sequential Social Dilemma is a partially observable Markov game $\mathcal{G} = \langle N, \mathcal{S}, \{\mathcal{A}_i\}_{i=1}^N, T, \{R_i\}_{i=1}^N, H \rangle$ with N agents, state space $\mathcal{S}$ (the gridworld configuration), per-agent action spaces $\mathcal{A}_i$, transition function T , reward functions R_i , and episode horizon H [1]. Beyond the dilemma’s matrix-game structure, SSDs add temporal richness: agents must learn *when* and *where* to cooperate, not just *whether* to.

We study two canonical SSDs that capture complementary dilemma types.

Cleanup [5] is a *public goods provision* game ($N=10$). The map has two regions: a river that accumulates waste, and an orchard where apples grow. Apples regrow only when river pollution is below threshold. Each agent can fire a cleaning beam (cost -1) that removes waste, collect apples ($+1$ each), or fire a tagging beam (cost -1 , inflicting -50 on the target and removing it for 25 steps). The dilemma: cleaning is costly but its benefits are public, so purely self-interested agents free-ride.

Gathering [1, 6] is a *common pool resource* game ($N=4$). Agents collect shared apples on a fixed respawn timer and may fire tagging beams to temporarily remove rivals. The dilemma: agents can coexist and share resources, or aggress to monopolize them; aggression wastes time and reduces total welfare.

Both games use 8–9 discrete actions (movement, rotation, beam, stand, optionally clean) and episodes of $H=1000$ steps. The two dilemmas differ in cost structure: asymmetric provision (cleaners pay, all benefit) vs. symmetric restraint (every agent faces the same temptation), a distinction that drives our experimental findings.

Social metrics. Following Perolat et al. [6], let $R_i = \sum_{t=0}^{H-1} r_i^t$ denote agent i ’s episode return. We evaluate four social outcomes:

$$U = \frac{1}{H} \sum_{i=1}^N R_i \quad (\text{efficiency}) \quad (1)$$

$$E = 1 - \frac{\sum_{i,j} |R_i - R_j|}{2N \sum_i R_i} \quad (\text{equality}) \quad (2)$$

$$S = \frac{1}{N} \sum_{i=1}^N \bar{t}_i \quad (\text{sustainability}) \quad (3)$$

$$P = \frac{1}{H} \sum_{t=0}^{H-1} |\{i : \text{active}_i^t\}| \quad (\text{peace}) \quad (4)$$

where $\bar{t}_i$ is the mean timestep at which agent i collects positive reward (higher $\Rightarrow$ resources preserved later) and active_i^t indicates that agent i is not tagged out at step t . We additionally consider the

76 *maximin* (Rawlsian) welfare criterion $\min_i R_i$, which optimizes the worst-off agent’s return and
 77 serves as the second objective in our experiments.

78 2.2 Iterative LLM Policy Synthesis

79 Let Π denote the space of *code-based policies*: deterministic functions $\pi : \mathcal{S} \times [N] \rightarrow \mathcal{A}$ expressed
 80 as executable Python code. Each policy has access to the full environment state and a library of
 81 helpers (BFS pathfinding, beam targeting, coordinate transforms). This state access is a deliberate
 82 design choice: programmatic policies operate in algorithm space rather than the reactive observation-
 83 to-action space of neural policies, which lets a single LLM generation step encode rich coordination
 84 logic.

85 A frozen LLM $\mathcal{M}$ acts as the *policy synthesizer*. Given a system prompt p describing the environ-
 86 ment API and a feedback prompt q_k , it produces a new policy

$$\pi_{k+1} = \mathcal{M}(p, q(\pi_k, \mathcal{F}_k)),$$

87 where π_k is the previous policy (its source code) and $\mathcal{F}_k$ is the evaluation feedback. All N agents
 88 execute the same program π_k in self-play. We stress that this is *symmetric*, not *behaviorally ho-*
 89 *mogeneous*: since π_k takes `agent_id` as an argument, a single shared program can induce distinct
 90 per-agent behaviors (cleaner vs. gatherer assignment, time-rotated duty cycles, partitioned territo-
 91 ries; see the synthesized policies in Appendix E). What is shared is the source code; the sampled
 92 action distributions can differ across agents. Evaluation over a set of random seeds S yields the
 93 mean per-agent return $\bar{r}_k$ and the social metrics vector $\mathbf{m}_k = (U_k, E_k, S_k, P_k)$. Each generated
 94 policy passes an AST-based safety check (blocking eval, file I/O, network access) followed by a
 95 short smoke test; failures trigger regeneration (up to R attempts) with the error message appended
 96 to the prompt.

97 **Feedback.** We package the previous policy’s source code together with all available evaluation
 98 signals:

$$\mathcal{F}_k = (\text{code}(\pi_k), \bar{r}_k, \mathbf{m}_k, \mathbf{d}), \quad (5)$$

99 where $\mathbf{d}$ contains natural-language definitions of each social metric. The LLM consumes $\mathcal{F}_k$ to
 100 revise and improve the policy. This is a starting point; the choice of feedback content is a single
 101 design decision within a much larger pipeline configuration space: our two-level framework (Sec-
 102 tion 3) opens the full space to automated search.

103 3 Two-Level Framework

104 We introduce a two-level system where a *researcher agent* $\mathcal{R}$ autonomously discovers configurations
 105 that optimize the output of an inner-loop system. While we instantiate this for multi-agent policy
 106 synthesis, the architecture is general: any pipeline where an LLM generates artifacts, evaluates them,
 107 and iterates can serve as the inner loop. The key insight is that the entire inner-loop codebase is a
 108 designable artifact that an AI agent can search over. Figure 1 illustrates the architecture.

109 3.1 Configuration Space

110 Let $\mathcal{C}$ denote the space of *pipeline configurations*. Each configuration $c \in \mathcal{C}$ specifies the full inner-
 111 loop setup:

$$c = (p, \phi, \mathcal{H}, \iota, v) \quad (6)$$

112 where p is the system prompt, ϕ is the feedback construction function (which metrics and diagnostics
 113 to include, how to frame it, whether to inject adaptive hints), $\mathcal{H}$ is the helper function library, ι
 114 specifies the iteration logic (number of iterations K , sampling strategy, temperature schedule), and
 115 v is the validation pipeline. Table 1 provides concrete examples.

116 The hand-designed feedback of [3] corresponds to a single fixed instantiation of ϕ . Our framework
 117 opens the full configuration space to automated search.

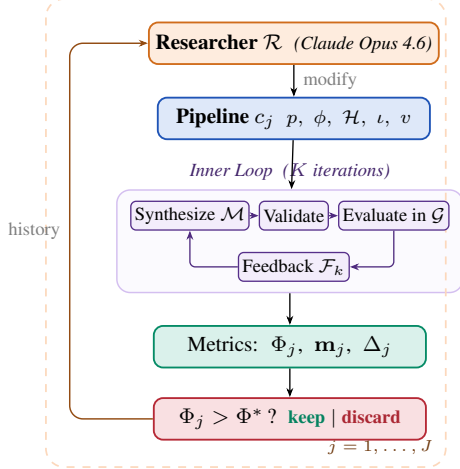


Figure 1: Two-level automated research framework (Algorithm 1).

Algorithm 1 Two-Level Automated Research

Require: Game $\mathcal{G}$, policy synthesizer $\mathcal{M}$, researcher $\mathcal{R}$, system prompt $p_{\mathcal{R}}$, initial config c_0 , outer iterations J , ground-truth objective Φ , held-out seeds S_{ho}

Ensure: Best configuration c^* , best policy π^*

```

1:  $\pi_0^* \leftarrow \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c_0)$ 
2:  $\Phi_0 \leftarrow \text{EVAL}(\pi_0^*; \mathcal{G}, S_{\text{ho}})$ 
3:  $\text{history} \leftarrow \{(c_0, \Phi_0, \mathbf{m}_0, \emptyset)\}$ 
4: for  $j = 1, \dots, J$  do
5:    $c_j \leftarrow \mathcal{R}(p_{\mathcal{R}}, \text{CODE}(c_{j-1}), \text{history})$ 
6:   if  $\neg \text{VALIDATECONFIG}(c_j)$  then  $\text{retry} \leq R$ 
7:      $\pi_j^* \leftarrow \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c_j)$ 
8:      $\Phi_j \leftarrow \text{EVAL}(\pi_j^*; \mathcal{G}, S_{\text{ho}})$ 
9:      $\Delta_j \leftarrow \text{DIFF}(c_{j-1}, c_j)$  // code diff
10:     $\text{history} \leftarrow \text{history} \cup \{(c_j, \Phi_j, \mathbf{m}_j, \Delta_j)\}$ 
11:  end for
12:  $j^* \leftarrow \arg \max_j \Phi_j$ 
13: return  $c_{j^*}, \pi_{j^*}^*$ 

```

Table 1: Configuration components modifiable by the researcher, with concrete edits that $\mathcal{R}$ made in our runs. The environment simulator, ground-truth evaluation, and policy LLM weights are frozen.

	Scope	Concrete edits by $\mathcal{R}$ (see appendix)
p	System prompt	Multi-section strategic briefing with explicit duty-rotation template (<code>agent_id + step//T</code>) % n (App. D.2, Listing 3)
ϕ	Feedback fn	Thresholded diagnostics: “FAIRNESS ALERT” fires when $\min_i R_i < 0$; “DO NOT REGRESS” guard fires when $U \geq 2.5$ (App. D.3, Listing 4)
$\mathcal{H}$	Helper library	BFS-Voronoi territories with respawn-timer awareness; band-based apple zoning by <code>agent_id</code> (App. D.1, Listings 2, 1)
ι	Iteration logic	Per-condition $K \in \{2, 3\}$; $ S $ walked $5 \rightarrow 8 \rightarrow 12$ for variance control; thinking budget 10–32k (App. D.4, Table 6)
v	Validation	AST blocks <code>eval/file I/O/network</code> ; 50-step smoke test; $\text{retry} \leq R$ on failure

3.2 Inner Loop (Policy Synthesis)

Given a configuration c , the inner loop executes K iterations of LLM policy synthesis:

$$\pi_c^* = \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c) \quad (7)$$

Each iteration k proceeds in four stages, following [3]:

- Synthesize.** The policy LLM $\mathcal{M}$ receives the system prompt p , the previous policy’s source code π_{k-1} , and feedback $\mathcal{F}_{k-1}$ constructed by ϕ . It generates a new Python policy function π_k that has access to full environment state and the helper library $\mathcal{H}$.
- Validate.** The generated code undergoes AST-based safety checks (blocking dangerous operations such as file I/O and network access) followed by a short smoke test. Failures trigger re-generation (up to R retries), with the error message appended to the prompt.
- Evaluate.** All N agents execute the same policy π_k in self-play over $|S|$ random seeds (note the policy is conditional on `agent_id`). The evaluation yields the mean per-agent reward $\bar{r}_k$ and the social metrics vector $\mathbf{m}_k = (U_k, E_k, S_k, P_k)$.
- Feedback.** The feedback function ϕ constructs the prompt for the next iteration from $(\pi_k, \bar{r}_k, \mathbf{m}_k)$, packaging the previous policy’s source code together with the scalar reward, the social metrics vector, their natural-language definitions, and any adaptive diagnostics that ϕ injects.

The inner loop output is evaluated on held-out seeds via a fixed ground-truth objective:

$$\Phi(c) = \text{EVAL}(\pi_c^*; \mathcal{G}, S_{\text{held-out}}) \quad (8)$$

135 where Φ maps from social outcomes to a scalar. We consider two objectives:

$$\Phi_U = U \quad (\text{utilitarian efficiency}) \quad (9)$$

$$\Phi_{\min} = \min_i R_i \quad (\text{Rawlsian maximin}) \quad (10)$$

136 3.3 Outer Loop (Automated Research)

137 The researcher agent $\mathcal{R}$ iteratively modifies the pipeline configuration. Following the autoresearch
138 paradigm [4], $\mathcal{R}$ operates on the inner-loop codebase as a modifiable artifact, proposing changes,
139 observing outcomes, and refining. The procedure is formalized in Algorithm 1.

140 At each outer iteration j , the researcher $\mathcal{R}$ receives: i) the **full source code** of the current config-
141 uration c_{j-1} (prompts, feedback construction, helpers, iteration logic); ii) the **experiment history**:
142 for each prior iteration, the code diff Δ_i , ground-truth score Φ_i , and social metrics vector $\mathbf{m}_i$; iii)
143 the **environment source code** (read-only), enabling the researcher to reason about game mechanics.
144 The researcher proposes a new configuration c_j by generating code modifications. Concretely, $\mathcal{R}$
145 is a coding agent (Claude Code CLI) that operates on a real software repository: it reads and edits
146 Python source files, runs shell commands, inspects evaluation outputs, and commits changes to a
147 dedicated git branch, following the same workflow a human researcher would follow. This mirrors
148 the original autoresearch setup [4], where an AI coding agent modifies `train.py` and observes
149 validation loss; here, the researcher modifies the inner-loop codebase and observes social welfare.

150 3.4 Connection to Automated Mechanism Design

151 The two-level structure admits a mechanism design interpretation. The researcher $\mathcal{R}$ acts as a *mech-*
152 *anism designer*: it controls the information structure (what metrics to reveal, how to frame them),
153 the action space (what helper functions are available), and the incentive structure (how feedback is
154 presented) under which the policy synthesizer $\mathcal{M}$ operates. The synthesizer acts as the *agent* within
155 the designed mechanism.

156 This connects to the automated mechanism design literature [7], where a principal designs rules to
157 induce desired behavior from self-interested agents. In our setting:

- 158 • The **principal** is the researcher $\mathcal{R}$, optimizing the ground-truth objective Φ .
- 159 • The **agent** is the synthesizer $\mathcal{M}$, optimizing per-agent reward as instructed.
- 160 • The **mechanism** is the configuration c : prompts, feedback, helpers, iteration logic.
- 161 • The **outcome** is the social welfare of the resulting multi-agent policy π_c^* .

162 A crucial distinction from classical mechanism design: $\mathcal{M}$ follows instructions but has *bounded*
163 *rationality* in the sense that its ability to synthesize effective policies depends on the information
164 and tools provided. The researcher’s task is thus closer to *information design* [8]: choosing what
165 to reveal to help $\mathcal{M}$ navigate the cooperation–defection tension. Our experiments (Section 4) show
166 that the researcher designs qualitatively different information structures depending on the welfare
167 objective Φ , supporting this interpretation empirically.

168 4 Experiments

169 4.1 Setup

170 We conduct 12 autonomous researcher runs across a factorial design. The researcher agent $\mathcal{R}$ is
171 Claude Opus 4.6, invoked via the Claude Code CLI as a coding agent. Each run operates on a dedi-
172 cated git branch of a real Python codebase: $\mathcal{R}$ edits source files in `pipeline/`, executes evaluation
173 scripts, reads metric outputs, and iterates, without human intervention.

174 **Design.** For Cleanup: 2 policy LLMs $\times$ 2 objectives $\times$ 2 replications = 8 runs. For Gathering: 2
175 policy LLMs $\times$ 1 objective $\times$ 2 replications = 4 runs. Maximin runs are unnecessary for Gather-
176 ing because efficiency optimization alone achieves equality $E > 0.94$ (the game’s symmetric cost
177 structure makes fairness a free byproduct of coordination).

178 **Models.** Policy synthesizer $\mathcal{M}$: Gemini 3.1 Pro (Google) or Claude Sonnet 4.6 (Anthropic). Both
179 use extended thinking. These are the same models evaluated in [3]. We additionally test with

Table 2: Results. Mean / max across 2 runs per condition (4 for Cleanup baselines). Baseline: unmodified pipeline from [3]. GEPA: automated prompt optimization [9] with matched compute budget.

Policy LLM	Target	U	E	$\min_i R_i$
<i>Cleanup (N=10) — Baseline</i>				
Gemini 3.1 Pro	—	1.93/2.70	0.17/0.62	−159/−84
Sonnet 4.6	—	0.86/1.56	−0.02/0.25	−151/−59
<i>Cleanup (N=10) — GEPA [9]</i>				
Gemini 3.1 Pro	Φ_U	1.34/1.37	−0.10/−0.05	−149/−126
	$\Phi_{\min}$	1.76/2.76	0.90/0.96	143/245
Sonnet 4.6	Φ_U	1.04/1.20	−0.59/−0.21	−164/−126
	$\Phi_{\min}$	−0.63/1.60	0.77/1.00	−147/6
<i>Cleanup (N=10) — Autoresearch</i>				
Gemini 3.1 Pro	Φ_U	3.20 /3.25	0.55/0.61	−211/−182
	$\Phi_{\min}$	3.16/3.19	0.98 /0.98	290 /296
Sonnet 4.6	Φ_U	3.12 /3.14	0.66/0.70	−196/−146
	$\Phi_{\min}$	2.57/2.93	0.91 /0.97	179 /200
<i>Gathering (N=4) — Baseline</i>				
Gemini 3.1 Pro	—	2.04/2.42	0.90/0.98	412/571
Sonnet 4.6	—	0.03/0.03	0.54/0.54	0/0
<i>Gathering (N=4) — GEPA [9]</i>				
Gemini 3.1 Pro	Φ_U	2.08/2.35	0.94/0.96	436/518
Sonnet 4.6	Φ_U	1.20/1.23	0.63/0.71	86/123
<i>Gathering (N=4) — Autoresearch</i>				
Gemini 3.1 Pro	Φ_U	2.49/2.51	0.98/0.98	582/582
Sonnet 4.6	Φ_U	2.52/2.52	0.96/0.98	576/597

180 Gemma 4 26B-A4B-IT (Google), a smaller open-weight model, to probe the framework’s behavior
 181 when the policy synthesizer has substantially lower capability (Appendix C).

182 **Baselines.** The hand-designed feedback configuration from [3] serves as the initial pipeline c_0
 183 for all runs. On Cleanup ($N=10$), this baseline achieves $U=1.93/2.70$ (mean/max) with Gemini
 184 and $U=0.86/1.56$ with Sonnet across 4 runs each. We additionally compare against GEPA [9], an
 185 automated prompt optimization method that iteratively refines the system prompt via LLM reflection.
 186 GEPA optimizes only the system prompt p , whereas our researcher modifies the full pipeline
 187 $(p, \phi, \mathcal{H}, \iota, v)$. We give GEPA a matched compute budget: the same number of optimization steps
 188 as outer iterations used by our method (each GEPA step costs ~ 3 policy evaluations, comparable to
 189 our inner loop’s $K+1=4$).

190 **Inner loop parameters.** $K=3$ synthesis iterations (or $K=2$ if the researcher elects), evaluated
 191 over $|S|=5-12$ random seeds (researcher-configurable). The researcher ran $J=3-17$ outer iterations
 192 per run (mean: 8.3), with a keep rate of 18%–100% (mean: 63%).

193 4.2 Results

194 Table 2 presents the main results, comparing autoresearch against the hand-designed baseline of [3]
 195 and GEPA prompt optimization [9].

196 **Finding 1: The researcher reliably improves over hand-designed baselines and outperforms**
 197 **prompt-only optimization.** All 12 runs show substantial improvement regardless of starting point
 198 (Figure 2). In Cleanup, the researcher exceeds the unmodified pipeline baseline by 66% with Gemini
 199 ($U: 1.93 \rightarrow 3.20$) and 263% with Sonnet ($U: 0.86 \rightarrow 3.12$). Strikingly, the Sonnet improvement
 200 nearly closes the gap with Gemini ($U=3.12$ vs. 3.20), despite a $2\times$ baseline disadvantage (0.86 vs.
 201 1.93). Final efficiency values cluster tightly around $U \approx 3.1-3.2$ despite baselines varying from
 202 0.29 to 2.70 across individual runs, suggesting the researcher reliably discovers the performance
 203 ceiling of each policy LLM. Beyond improving the mean, the researcher yields remarkably consistent
 204 results: under efficiency optimization, Gemini’s runs cluster within $U=3.20-3.25$ (gap 0.05)

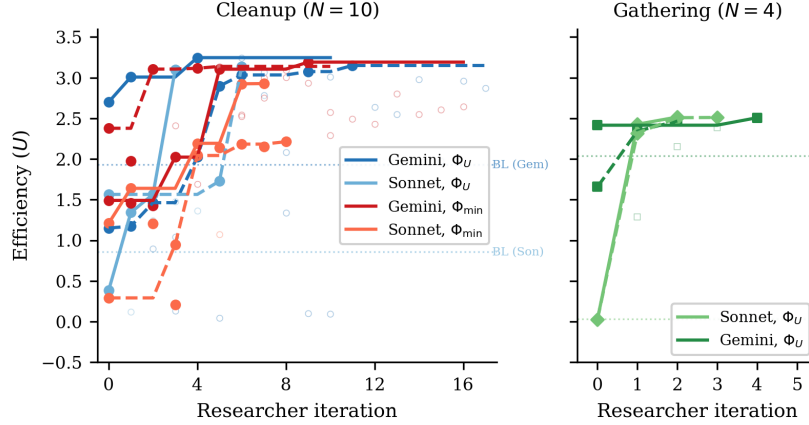


Figure 2: Efficiency (U) across researcher iterations for all 12 runs. *Left*: Cleanup ($N=10$) with 8 runs across 2 LLMs $\times$ 2 objectives (solid lines = kept experiments, open circles = discarded). Dashed horizontal line: hand-designed baseline from [3]. All runs converge to $U \approx 3.1$ – 3.2 despite diverse starting points. *Right*: Gathering ($N=4$) with 4 runs (2 per LLM). All converge to $U \approx 2.5$ within 2–4 iterations despite baselines ranging from 0.03 to 2.42.

despite baselines spanning 1.15–2.70, and Sonnet’s within 3.12–3.14 (gap 0.02) from baselines of 0.38–1.56. In Gathering, all four runs converge to $U=2.47$ – 2.52 from baselines ranging 0.03–2.42. The researcher discovers pipeline modifications: structured helpers (App. D.1), strategic hints (App. D.2), tailored feedback (App. D.3). These constrain the policy LLM toward consistently good solutions.

Compared to GEPA (Table 2), autoresearch achieves $2.4\times$ higher efficiency under Φ_U in Cleanup ($U=3.20$ vs. 1.34), $2\times$ higher maximin under $\Phi_{\min}$ (290 vs. 143), and 20% higher efficiency in Gathering ($U=2.49$ vs. 2.08). Critically, GEPA shows far wider spread across runs on the harder fairness objective ($\Phi_{\min}$ maximin in Cleanup: 245/41 vs. autoresearch’s 296/284). The gap widens further with Sonnet: on Cleanup Φ_U , GEPA stalls at $U=1.04$ ($3\times$ below autoresearch’s 3.12), and on $\Phi_{\min}$ its two runs split between a modestly positive solution ($\min_i R_i=6$, $U=1.60$) and a pathological “everyone cleans, nobody eats” regime (perfect equality $E=1.00$ but negative efficiency $U=-2.87$ and $\min_i R_i=-299$), while autoresearch with the same model reaches $\min_i R_i=179/200$ reliably. This highlights the advantage of modifying the full pipeline, whereas prompt-only optimization cannot address the underlying stochasticity of code generation.

Finding 2: No efficiency–fairness tradeoff in Cleanup (Gemini). Maximin-optimized Gemini pipelines sacrifice only 1% efficiency (U : 3.16 vs. 3.20) while achieving near-perfect equality (E : 0.98 vs. 0.55) and transforming maximin from deeply negative baselines (−99 to −84) to $\min_i R_i = 290$. The researcher discovers *fair duty rotation* (Listing 6; primed by the prompt rewrite of Listing 3)—time-based cycling using `agent_id` and `env._step_count`—simultaneously improving worst-off welfare *and* collective output. Because cleaning is a public good, distributing the cleaning cost fairly ensures enough cleaners to sustain apple production.

For Sonnet, there is a moderate tradeoff: efficiency drops from 3.12 to 2.57 under maximin optimization. The gap reflects Sonnet’s harder time implementing complex coordination mechanisms (role rotation, zone assignment) from strategic hints alone.

Finding 3: Game structure determines whether fairness requires explicit optimization. In Cleanup, where cleaning costs are borne asymmetrically (cleaners pay −1, free-riders collect apples), baseline equality ranges from $E=0.04$ to 0.62, and maximin optimization is required to reach $E > 0.9$. In Gathering, where all agents face a symmetric landscape, efficiency optimization alone achieves $E > 0.94$ across all 4 runs: no separate maximin runs are needed.

This generalizes: in *provision* dilemmas with asymmetric costs, fairness requires designed mechanisms (role rotation, duty sharing); in *restraint* dilemmas with symmetric costs, fairness emerges as a

free byproduct of efficient coordination. The researcher independently discovers this, it creates role differentiation pipelines only for Cleanup, and pure spatial-coordination pipelines for Gathering.

Finding 4: Convergent discovery of qualitatively different strategies per objective. Despite fully independent runs, the researcher converges on the same core strategies within each condition (Table 3, Appendix A). In Cleanup, waste-counting helpers and spatial zone partitioning appear across all 8 runs. The key qualitative difference is *role rotation*, appearing in 4/4 maximin runs (Listings 6, 7) but 0/4 efficiency runs (Listing 5). Under efficiency optimization, the researcher assigns static cleaning roles (some agents always clean), achieving high collective output at the cost of equality. Under maximin optimization, it discovers rotation as a fairness mechanism.

In Gathering, the researcher discovers BFS-Voronoi territory partitioning and respawn-timer awareness (Listings 2, 8), with no role differentiation. Thus, optimization is purely spatial (who collects which apples) and temporal (respawn-aware positioning).

Common failure modes. Several patterns led to discarded experiments across all runs: (1) *over-prescription*: adding too many strategic hints confuses the policy LLM, causing generation failures; (2) *iteration regression*: $K=4$ or $K=5$ inner iterations often cause catastrophic regression as the LLM “over-refines” a working policy; (3) *feedback overload*: verbose per-agent diagnostics cause over-correction rather than gradual improvement; (4) *variance exposure*: identical configurations can yield $U=3.2$ then $U=0.09$ on different seeds, requiring multi-seed evaluation for stable signals (per-run K , $|S|$ in Table 6).

5 Related Work

Sequential social dilemmas. SSDs were introduced by Leibo et al. [1] (Gathering) and extended to public-goods settings by Hughes et al. [5] (Cleanup); Perolat et al. [6] formalized the social outcome metrics (U, E, S, P) used here. Subsequent work studies inequity aversion, intrinsic motivation, and reputational mechanisms for promoting cooperation in MARL agents. We complement this line by automating the search for cooperative *programs* rather than evolving neural policies, and by treating the welfare objective Φ as a designable parameter that the outer agent optimizes for, obtaining qualitatively different cooperative behaviors (static role assignment vs. duty rotation) under different objectives without changing the environment.

LLMs for policy and program synthesis. FunSearch [10] evolves programs for combinatorial discovery; Eureka [11] synthesizes reward functions from environment source code; Voyager [12] and Code as Policies [13] generate executable skill code for single-agent embodied control; ReEvo [14] evolves heuristics with reflective feedback. These works target single-agent settings or non-strategic optimization. Gallego [3], on which our inner loop is based, extended LLM program synthesis to the multi-agent SSD setting, where one program must coordinate N self-play copies. Our contribution is one level above: rather than tuning the synthesizer for one task, we let an outer agent rewrite the synthesis pipeline itself.

LLM reflection and prompt optimization. Reflexion [15], Self-Refine [16], OPRO [17], and GEPA [9] demonstrate that structured verbal feedback loops improve LLM outputs; ERL [18] internalizes such reflection via self-distillation. These methods optimize the prompt or the model’s reasoning trajectory in isolation. Our framework instead modifies the entire surrounding pipeline (system prompt, feedback construction, helper library, iteration logic, validation) making prompt optimization one component of a strictly larger search space. The empirical gap is large: Section 4 shows full-pipeline autoresearch achieving $3\times$ higher efficiency in Cleanup and 37% higher in Gathering than GEPA at matched compute, with run-to-run spread an order of magnitude tighter on the harder fairness objective.

Automated AI research. A small but growing body of work delegates the design of ML pipelines to LLM coding agents. Karpathy’s autoresearch [4] runs a coding agent that modifies `train.py` for nanoGPT pretraining and is rewarded for validation loss. Our system shares this autoresearch architecture (coding agent + frozen evaluation harness + diff-based history) but targets a different inner loop (multi-agent policy synthesis instead of single-model pretraining) and a different objective

(multi-agent social welfare instead of validation bits-per-byte). To our knowledge this is the first instantiation of the autoresearch paradigm in a multi-agent decision-making domain.

Automated mechanism and information design. Classical automated mechanism design [7] computes optimal allocation rules for self-interested agents under explicit incentive constraints, while information design [8] studies how a principal commits to a signaling policy that shapes a receiver’s behavior. Our outer agent solves a related but distinct problem: $\mathcal{M}$ is not strategically deceptive (it follows instructions) but is *boundedly rational*, so the researcher must decide what information, helpers, and structure to expose so that $\mathcal{M}$ writes policies achieving the principal’s welfare goal. Section 4 shows that the pipelines $\mathcal{R}$ produces are genuinely a function of the welfare objective, supporting this information-design framing empirically.

6 Discussion and Conclusion

We have presented a two-level framework where an AI coding agent autonomously discovers pipeline configurations that improve the output of an inner-loop LLM system. Applied to multi-agent policy synthesis in social dilemmas, the researcher agent reliably exceeds hand-designed baselines across 12 independent runs, and converges on qualitatively similar strategies within each game-objective condition. The framework requires no domain-specific scaffolding: the agent operates on a standard software repository using the same tools (file editing, shell commands, git) available to a human researcher.

Mechanism design in action. Our results empirically support the mechanism design interpretation of Section 3.4. The researcher acts as an information designer: under Φ_U , it reveals efficiency-oriented information (waste counts, zone assignments) that guides $\mathcal{M}$ toward productive but unequal role allocation (Listing 5). Under $\Phi_{\min}$, it additionally reveals fairness-oriented structure such as rotation schedules and per-agent equity feedback (Listings 3, 4), guiding $\mathcal{M}$ toward egalitarian coordination. The striking absence of a fairness–efficiency tradeoff with Gemini suggests that in public goods provision, the mechanism design problem has a cooperative optimum where both objectives align.

Two types of dilemma. The Cleanup–Gathering contrast reveals a structural insight about social dilemmas: *provision* dilemmas (asymmetric costs, one agent’s contribution benefits all) require explicit fairness mechanisms, while *restraint* dilemmas (symmetric costs, each agent faces the same temptation) achieve fairness naturally when coordination improves. This distinction, well-known in the common-pool resource literature, is rediscovered here by an autonomous AI agent, which suggests it is a robust structural feature of these environments.

Human oversight and audit. Our system is fully autonomous by design, but its architecture offers natural affordances for human-in-the-loop oversight. Because the researcher $\mathcal{R}$ operates on a standard git repository, a domain expert can audit the full history of code diffs Δ_j alongside their metric outcomes $(\Phi_j, \mathbf{m}_j)$, exactly as they would review a colleague’s pull requests. The keep/discard decision (Algorithm 1, line 11) is a lightweight intervention point: a human could override the accept threshold, veto modifications that introduce undesirable mechanisms (e.g., exploitative role assignments), or steer the researcher toward under-explored regions of the configuration space by editing the experiment history. More broadly, the separation between the researcher $\mathcal{R}$ and the frozen evaluation Φ creates a *delegation boundary*: the human defines *what* to optimize (the welfare objective), while the agent decides *how*. This mirrors the expert-in-the-loop design patterns identified in deployment-oriented work on AI agents for discovery, where trust calibration depends on the legibility of the agent’s decision trail rather than on real-time supervision.

Future work. Several directions emerge. First, we are interested in applying the two-level framework to other LLM-driven pipelines (code optimization, scientific experiment design, infrastructure tuning), where the same architecture applies with a different inner loop and objective Φ ; next, adversarial objectives can be intriguing, to test whether the researcher discovers exploitative pipeline configurations, exposing reward hacking risks; and asymmetric programs: extend to settings where agents run *different* source code (the present setup is symmetric in code but already supports hetero-

337 geneous behavior via `agent_id`; relaxing the shared-program constraint enables richer per-agent
338 specialization and partial-information settings).

339 References

- 340 [1] Joel Z Leibo, Vinicius Zambaldi, Marc Lanctot, Janusz Marecki, and Thore Graepel. Multi-
341 agent reinforcement learning in sequential social dilemmas. In *Proc. 16th Conference on Au-*
342 *tonomous Agents and MultiAgent Systems*, pages 464–473, 2017.
- 343 [2] Lucian Buşoniu, Robert Babuška, and Bart De Schutter. A comprehensive survey of multia-
344 gent reinforcement learning. *IEEE Transactions on Systems, Man, and Cybernetics, Part C*
345 *(Applications and Reviews)*, 38(2):156–172, 2008.
- 346 [3] Víctor Gallego. Cooperation and exploitation in LLM policy synthesis for sequential social
347 dilemmas. In *Proc. Adaptive and Learning Agents Workshop (ALA 2026)*, 2026.
- 348 [4] Andrej Karpathy. autoresearch: AI agents running research on single-GPU nanochat training
349 automatically. <https://github.com/karpathy/autoresearch>, 2026.
- 350 [5] Edward Hughes, Joel Z Leibo, Matthew Phillips, Karl Tuyls, Edgar A Dueñez-Guzmán, Anto-
351 nio García Castañeda, Iain Dunning, Tina Zhu, Kevin R McKee, R. Koster, Heather Roff, and
352 Thore Graepel. Inequity aversion improves cooperation in intertemporal social dilemmas. In
353 *Neural Information Processing Systems*, 2018.
- 354 [6] Julien Perolat, Joel Z Leibo, Vinicius Zambaldi, Charles Beattie, Karl Tuyls, and Thore Grae-
355 pel. A multi-agent reinforcement learning model of common-pool resource appropriation. In
356 *Advances in Neural Information Processing Systems*, volume 30, 2017.
- 357 [7] Vincent Conitzer and Tuomas Sandholm. Complexity of mechanism design. In *Proc. 18th*
358 *Conference on Uncertainty in Artificial Intelligence*, pages 103–110, 2002.
- 359 [8] Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*,
360 101(6):2590–2615, 2011.
- 361 [9] Lakshya A Agrawal et al. GEPA: Reflective prompt evolution can outperform reinforcement
362 learning. *arXiv preprint arXiv:2507.19457*, 2026.
- 363 [10] Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog,
364 M Pawan Kumar, Emilien Dupont, Francisco JR Ruiz, Jordan S Ellenberg, Pengming Wang,
365 Omar Fawzi, et al. Mathematical discoveries from program search with large language models.
366 *Nature*, 625(7995):468–475, 2024.
- 367 [11] Yecheng Jason Ma, William Liang, Guanzhi Wang, De-An Huang, Osbert Bastani, Dinesh
368 Jayaraman, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Eureka: Human-level reward
369 design via coding large language models. In *The Twelfth International Conference on Learning*
370 *Representations*, 2024.
- 371 [12] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi
372 Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language
373 models. *Transactions on Machine Learning Research*, 2024.
- 374 [13] Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence,
375 and Andy Zeng. Code as policies: Language model programs for embodied control. In *2023*
376 *IEEE International Conference on Robotics and Automation (ICRA)*, pages 9493–9500, 2023.
- 377 [14] Haoran Ye, Jiarui Wang, Zhiguang Cao, Federico Berto, Chuanbo Hua, Haeyeon Kim, Jinkyoo
378 Park, and Guojie Song. ReEvo: Large language models as hyper-heuristics with reflective
379 evolution. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*,
380 2024.
- 381 [15] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Re-
382 flexion: language agents with verbal reinforcement learning. In *Thirty-seventh Conference on*
383 *Neural Information Processing Systems*, 2023.

- [16] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. Self-refine: Iterative refinement with self-feedback. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.
- [17] Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V Le, Denny Zhou, and Xinyun Chen. Large language models as optimizers. In *The Twelfth International Conference on Learning Representations*, 2024.
- [18] Taiwei Shi, Sihao Chen, Bowen Jiang, Linxin Song, Longqi Yang, and Jieyu Zhao. Experiential reinforcement learning. *arXiv preprint arXiv:2602.13949*, 2026.

A Additional Results

Table 3: Strategies discovered by the researcher in Cleanup across 4 efficiency-optimized and 4 maximin-optimized independent runs.

Strategy	Φ_U (4 runs)	$\Phi_{\min}$ (4 runs)
Waste-counting helpers	4/4	4/4
Zone/lane partitioning	3/4	4/4
Anti-regression feedback	3/4	2/4
Worked policy examples	2/4	3/4
Cleaning cost economics	2/4	3/4
Time-based role rotation	0/4	4/4

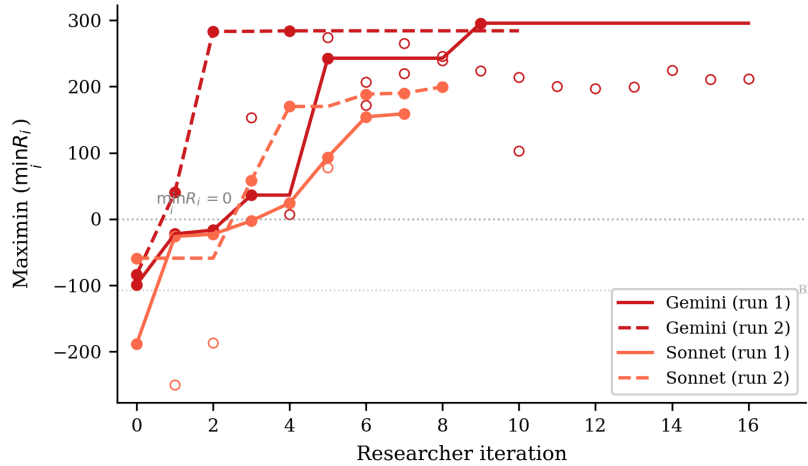


Figure 3: Maximin ($\min_i R_i$) across researcher iterations for the 4 Cleanup maximin-optimized runs. All runs transform deeply negative baselines (worst-off agents losing reward) into substantially positive values. Gemini reaches ~ 290 while Sonnet reaches ~ 160 – 200 . The dashed line marks $\min_i R_i = 0$ (no agent loses reward overall).

B Compute Requirements

Table 4 reports wall-clock time for all 12 autonomous runs and the 6 GEPA comparison runs. *Inner loop* measures total time spent on policy LLM generation and environment simulation across all evaluations; *total* (estimated from run-directory timestamps) additionally includes the researcher agent’s analysis, code editing, and decision-making between evaluations.

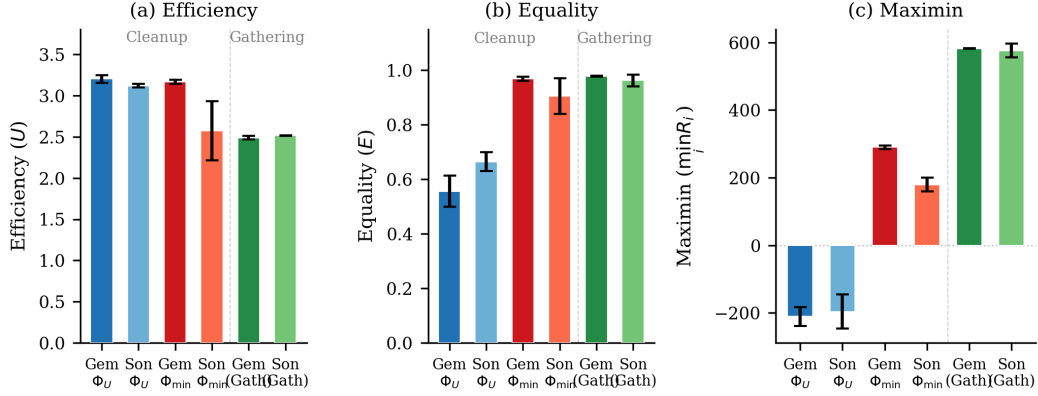


Figure 4: Final metrics across all conditions. (a) Efficiency: all conditions converge to $U \approx 2.5$ – 3.2 . (b) Equality: maximin-optimized Cleanup runs achieve $E \approx 1.0$, while efficiency-optimized runs show $E \approx 0.5$ – 0.7 ; Gathering achieves high equality regardless. (c) Maximin: the sharpest contrast—efficiency optimization leaves the worst-off agent deeply negative ($\min_i R_i \approx -200$), while maximin optimization transforms it to $+290$ (Gemini) or $+179$ (Sonnet). Gathering achieves high maximin (~ 580) under efficiency optimization alone. Error bars show s.d. across runs.

Table 4: Wall-clock time per autonomous run. Evals: total inner loop evaluations including initial baseline. Per eval: mean wall-clock per single evaluation. Total: end-to-end including researcher overhead.

Policy LLM	Φ	Evals	Inner loop (h)	Per eval (min)	Total (h)
<i>Cleanup ($N=10$)</i>					
Gemini	Φ_U	11	2.6	14	3.7
Gemini	Φ_U	18	4.1	14	6.4
Sonnet	Φ_U	4	2.1	32	6.1
Sonnet	Φ_U	7	5.0	43	6.1
Gemini	$\Phi_{\min}$	17	3.0	11	4.3
Gemini	$\Phi_{\min}$	11	3.6	20	5.0
Sonnet	$\Phi_{\min}$	8	4.5	33	8.8
Sonnet	$\Phi_{\min}$	9	5.3	36	13.2
<i>Gathering ($N=4$)</i>					
Sonnet	Φ_U	3	1.7	33	2.2
Gemini	Φ_U	5	1.4	17	1.9
Gemini	Φ_U	3	0.9	19	1.1
Sonnet	Φ_U	4	2.3	35	3.4
All 12 runs		100	36.6	22	62.2
GEPA (6 runs)		6	5.0	50	—

Cost breakdown. Each inner loop evaluation involves $K=2$ – 3 policy LLM generation calls (each $\sim 2k$ input tokens, $\sim 1.5k$ output tokens) followed by multi-seed simulation (5–12 seeds, ~ 15 – $30s$ total). Policy LLM generation dominates inner loop time (86–97%), with Sonnet evaluations taking $\sim 2\times$ longer than Gemini due to extended thinking. The researcher agent (Claude Opus 4.6, running via Claude Code CLI) accounts for 41% of total wall-clock time (25.6h of the 62.2h total across all 12 runs), spent reading results, editing pipeline source files, and planning modifications. In monetary terms, the researcher agent is the dominant cost: each outer iteration consumes ~ 50 – $100k$ context tokens in the Opus session, whereas each inner loop evaluation uses only ~ 5 – $10k$ policy LLM tokens. The 6 GEPA comparison runs required 5.0h of compute with no researcher agent, operating at lower cost per run but achieving substantially lower performance (Table 2).

C Smaller Open-Weight Model: Gemma 4 26B

We test the framework with Gemma 4 26B-A4B-IT (Google), a 26B-parameter open-weight model substantially smaller than the frontier models in the main experiments. We run three experiments—two on Cleanup ($N=10$) optimizing efficiency and maximin respectively, and one on Gathering ($N=4$) optimizing efficiency—all with Opus 4.6 as the researcher $\mathcal{R}$.

Setup. In all three experiments, Gemma starts from complete failure: the baseline pipeline produces $U=-10.0$ on Cleanup (agents CLEAN-spam without collecting apples) and $U=0.0$ on Gathering (broken BFS calls), compared to $U=1.93/0.86$ (Gemini/Sonnet on Cleanup) and $U=2.04/0.03$ (Gemini/Sonnet on Gathering). The researcher ran $J=10-18$ outer iterations per condition.

Results. Table 5 presents the best configurations discovered. Under efficiency optimization, the researcher achieves $U=0.87$ —a dramatic recovery from the broken baseline but far below frontier models ($U \approx 3.2$). The researcher compensates for the model’s limited capability by reducing inner-loop iterations to $K=1$ (avoiding the regression that plagued $K \geq 2$ runs) and providing highly structured worked examples with explicit cleaning-role assignment.

Table 5: Autoresearch with Gemma 4 26B-A4B-IT. Single run per condition. Baseline: pipeline from [3].

Game	Target	U	E	$\min_i R_i$	Keep rate
Cleanup	Baseline	-10.0	1.00 [†]	-1000	—
	Φ_U	0.87	-1.11	-371	6/19
	$\Phi_{\min}$	1.71	0.94	137	9/17
Gathering	Baseline	0.0	1.00 [†]	0	—
	Φ_U	2.44	0.98	580	4/11

[†]Equality is trivially 1.0 because all agents receive near-zero reward.

Maximin optimization rescues efficiency. Strikingly, under maximin optimization the researcher achieves *higher* efficiency ($U=1.71$) than under direct efficiency optimization ($U=0.87$), alongside strong equality ($E=0.94$) and positive worst-off welfare ($\min_i R_i=137$). This reversal—absent in frontier models, where both objectives yield $U \approx 3.1-3.2$ —occurs because the maximin objective forces the researcher toward coordination mechanisms (rotating cleaning duties, index-based apple assignment) that simultaneously improve collective output. Under efficiency-only optimization, the researcher converges on a local optimum—static role assignment with 25% dedicated cleaners—that a 26B model can implement but that caps performance well below the frontier.

This suggests that *for models below a capability threshold, fairness objectives may serve as better optimization targets for overall social welfare than direct efficiency maximization*. The structured coordination enforced by the maximin objective provides a scaffold that compensates for the weaker model’s difficulty implementing complex strategies from strategic hints alone.

Gathering: full recovery on a simpler game. On Gathering ($N=4$), the researcher brings Gemma from $U=0.0$ to $U=2.44$ with $E=0.98$ and $\min_i R_i=580$, after 10 outer iterations (4 kept). These values nearly match frontier models (Gemini: $U=2.49$, Sonnet: $U=2.52$; Table 2). The researcher discovers the same Voronoi partitioning and respawn-aware camping strategies found in the main experiments, and increases inner-loop iterations to $K=5$ to allow sufficient refinement. The contrast with Cleanup—where Gemma peaks at $U=0.87/1.71$ versus frontier $U \approx 3.2$ —suggests that the researcher can fully compensate for model weakness on simpler coordination tasks (4 agents, symmetric costs) but not on harder provision dilemmas (10 agents, asymmetric costs).

D Researcher-Authored Pipeline Artifacts

The previous appendix shows the policies π_c^* that the inner loop *outputs*; this one shows the configuration $c = (p, \phi, \mathcal{H}, \iota, v)$ (Section 4, Table 1) that the researcher $\mathcal{R}$ *authored* to produce them. Excerpts are taken from the final commit on the dedicated git branch of the corresponding run; we

447 reproduce the prose verbatim, with author commentary in [brackets] and elisions marked
 448 We organize by artifact type so each subsection makes a within-type contrast (e.g., Φ_U vs. $\Phi_{\min}$, or
 449 Cleanup vs. Gathering).

450 D.1 Helper library $\mathcal{H}$: coordination primitives

451 The researcher adds primitives that the policy LLM can call as black boxes, sparing it from re-
 452 implementing tricky logic on every iteration. Two patterns dominate: (i) *state inspection* (waste
 453 counts, fractions, beam-yield scoring) and (ii) *coordination primitives* (zone assignment, role rota-
 454 tion). We show one of each.

Listing 1: Cleanup, $\Phi_{\min}$ – band-based apple zoning helper added by the researcher in exp5. This is the primitive that lets the policy in Listing 6 keep collectors within their own row band, preventing all 7 gatherers from racing to the same apple.

```

455 1 def get_my_apples(env, agent_id):
456 2     """Alive apples in this agent's horizontal row band.
457 3     Divides the orchard into n_agents bands; falls back to all apples if empty."""
458 4     aid = int(agent_id)
459 5     n = int(env.n_agents)
460 6     band_h = env.height / n
461 7     band_lo, band_hi = aid * band_h, (aid + 1) * band_h
462 8
463 9     my_apples, all_apples = set(), set()
464 10    for i in range(env.n_apples):
465 11        if env.apple_alive[i]:
466 12            ar = int(env._apple_pos[i, 0]); ac = int(env._apple_pos[i, 1])
467 13            all_apples.add((ar, ac))
468 14            if band_lo <= ar < band_hi:
469 15                my_apples.add((ar, ac))
470 16    return my_apples if my_apples else all_apples
471 17

```

Listing 2: Gathering – BFS-Voronoi territory + respawn-aware waiting helpers added by the researcher in gather-exp3. These are the primitives that Listing 8 composes into a one-line policy: “go to my_zone_apples, otherwise nearest_respawning_apple.”

```

473 1 def voronoi_zones(env):
474 2     """Multi-source BFS Voronoi over walkable cells.
475 3     Returns (row, col) -> agent_id; ties broken by lower agent_id."""
476 4     queue, visited = deque(), {}
477 5     for a_id in range(env.n_agents):
478 6         if int(env.agent_timeout[a_id]) == 0:
479 7             ar = int(env.agent_pos[a_id][0]); ac = int(env.agent_pos[a_id][1])
480 8             queue.append((ar, ac, a_id))
481 9             visited[(ar, ac)] = a_id
482 10    while queue:
483 11        r, c, a_id = queue.popleft()
484 12        for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
485 13            nr, nc = r + dr, c + dc
486 14            if 0 <= nr < env.height and 0 <= nc < env.width \
487 15                and not env.walls[nr, nc] and (nr, nc) not in visited:
488 16                visited[(nr, nc)] = a_id
489 17                queue.append((nr, nc, a_id))
490 18    return visited
491 19
492 20 def nearest_respawning_apple(env, agent_id, zones=None, max_timer=10):
493 21     """Nearest dead apple in MY zone respawning within max_timer steps.
494 22     Returns (row, col) of best wait spot, or None."""
495 23     if zones is None: zones = voronoi_zones(env)
496 24     best_pos, best_t, best_d = None, max_timer + 1, float('inf')
497 25     ar = int(env.agent_pos[agent_id][0]); ac = int(env.agent_pos[agent_id][1])
498 26     for i in range(env.n_apples):
499 27         if not env.apple_alive[i]:
500 28             pos = (int(env._apple_pos[i][0]), int(env._apple_pos[i][1]))
501 29             if zones.get(pos) == agent_id:
502 30                 t = int(env.apple_timer[i])
503 31                 if t <= max_timer:
504 32                     d = abs(pos[0] - ar) + abs(pos[1] - ac)
505 33                     if t < best_t or (t == best_t and d < best_d):
506 34                         best_pos, best_t, best_d = pos, t, d
507 35    return best_pos
508 36

```

510 The choice of helper depends on the dilemma’s geometry: row-bands suffice when fairness is the
 511 main concern (Cleanup maximin) but full BFS-Voronoi is needed when wall-aware territory owner-

512 ship matters (Gathering). The researcher does not fix this in advance — it picks the simpler primitive
 513 whenever it works.

514 D.2 System prompt p : from neutral framing to strategic briefing

515 The unmodified system prompt is a neutral API description: “Write a policy that maximizes per-
 516 agent reward.” Under maximin optimization, the researcher rewrites the prompt into a multi-section
 517 strategic briefing. The excerpt below shows the pieces it added to the Cleanup prompt in exp5; the
 518 full prompt grew from 165 to 325 lines.

Listing 3: Cleanup, $\Phi_{\min}$ – excerpts from the researcher-rewritten system prompt (exp5). The unmodified baseline contained only API documentation; everything below was added by $\mathcal{R}$.

```

519 ## CRITICAL OBJECTIVE: Rawlsian Fairness (Maximin)
520
521 Your goal is to maximize the minimum per-agent total return across all
522 agents. This is the "maximin" or Rawlsian welfare criterion.
523
524 This means: it is NOT enough for the *average* agent to do well. The
525 worst-off agent must do as well as possible. A policy where 8 agents
526 each earn +200 but 2 agents each earn -100 is TERRIBLE (maximin = -100).
527
528 Key implication: cleaning costs (-1 per CLEAN action) must be shared
529 equitably among ALL agents. If you assign fixed "cleaner" roles, those
530 agents will accumulate large negative rewards and destroy your maximin score.
531
532 ## Strategy for Maximin: ROLE ROTATION + APPLE ZONING
533
534 The optimal strategy for maximin involves:
535 1. Shared cleaning duty: ALL agents take turns cleaning using a rotation
536    schedule based on 'agent_id' and 'env._step_count'. For example:
537    'is_my_cleaning_turn = (agent_id + env._step_count // SHIFT) % env.n_agents < NUM_CLEANERS'
538    where SHIFT is ~50 steps and NUM_CLEANERS is 2-3.
539 2. When NOT your turn: collect apples from YOUR ZONE using
540    'get_my_apples(env, agent_id)' -- prevents all gatherers competing
541    for the same nearest apple.
542 3. NEVER use BEAM (action 6): it costs -1 and causes -50 to the target.
543 4. Keep waste density low: aim for ~2-3 active cleaners at any time.
544
545 [... API documentation, helper documentation, working example ...]
546
547 IMPORTANT:
548 - NEVER assign permanent cleaning roles to specific agents -- this kills maximin.
549 - Use env._step_count for time-based role rotation so all agents share cleaning duty.
550 - NEVER use BEAM (action 6) -- it destroys both agents' rewards.
551

```

553 Two things stand out. First, the researcher writes the formula it expects $\mathcal{M}$ to use almost verbatim —
 554 $(\text{agent_id} + \text{env}._step_count // \text{SHIFT}) \% \text{env}._n_agents < \text{NUM_CLEANERS}$ — and the
 555 policy in Listing 6 uses essentially this template. Second, the researcher *teaches by counter-example*:
 556 the explicit “8 agents earn +200, 2 earn -100, maximin = -100” makes the failure mode of Φ_U -style
 557 static roles (Listing 5) concretely visible to $\mathcal{M}$. This is the information-design move predicted by
 558 the mechanism-design framing in Section 3.4.

559 The Gathering prompt evolves along a different axis (no maximin, no rotation). Its researcher-added
 560 “Key Strategic Insights” section instead emphasizes (i) *self-play implies never beam* and (ii) *Voronoi*
 561 *partition each step*, exactly matching what the policy in Listing 8 implements.

562 D.3 Feedback ϕ : adaptive diagnostics

563 The baseline feedback function from [3] shows reward + four social metrics with definitions and
 564 stops there. The researcher turns it into a state machine: it inspects the latest metrics and condi-
 565 tionally injects different hints. Two different runs produced two qualitatively different diagnostics
 566 — the same metric ($\bar{r}_k$ or $\min_i R_i$) is repurposed to either *stabilize* a working policy or *redirect* a
 567 failing one.

Listing 4: Adaptive feedback diagnostics. *Top*: efficiency-optimized run (exp1) – a stability guard prevents iter-3 regression once U is high. *Bottom*: maximin-optimized run (exp5) – two fairness diagnostics that fire when the worst-off agent loses reward.

```

568 1 # --- Cleanup, Phi_U feedback (exp1, pipeline/feedback.py final) ---
569 2 last_eff = history[-1]["metrics"]["efficiency"]
570

```

```

571 3 if last_eff >= 2.5:
572 4     parts.append(
573 5         """**CRITICAL -- DO NOT REGRESS**: The current policy achieves high "
574 6         "efficiency (>=2.5). You MUST output a policy that is nearly identical "
575 7         "to the current one. Copy the current policy code and make AT MOST "
576 8         "one small targeted improvement (e.g., adjust a single numeric "
577 9         "threshold). If you are not confident a change will help, output the "
578 10        "current policy UNCHANGED. A regression here means the run fails.")
579 11
580 12 # --- Cleanup, Phi_min feedback (exp5, pipeline/feedback.py final) ---
581 13 last_maximin = history[-1]["metrics"].get("maximin", 0)
582 14 last_avg      = history[-1]["reward_avg"]
583 15 if last_maximin < 0:
584 16     parts.append(
585 17         f"""**FAIRNESS ALERT**: maximin={last_maximin:.1f} is NEGATIVE. "
586 18         f"The worst-off agent lost reward overall while average was {last_avg:.1f}. "
587 19         "This means cleaning duties are NOT shared equitably. "
588 20         "Use time-based role rotation (env._step_count) so ALL agents share "
589 21         "cleaning costs equally. NEVER assign permanent cleaner roles.")
590 22 elif last_maximin < last_avg * 0.5:
591 23     parts.append(
592 24         f"""**FAIRNESS WARNING**: maximin={last_maximin:.1f} is much lower than "
593 25         f"average={last_avg:.1f}. The gap suggests unequal cleaning burden. "
594 26         "Ensure ALL agents rotate through cleaning duty.")

```

596 The two diagnostics are written by independent researcher runs but converge on a common pattern:
 597 *thresholded interventions on a primary metric*. They are also a common cause of the low run-to-
 598 run variance reported in Finding 1: when a working policy lands in the high- U basin, the stability
 599 guard prevents the LLM from over-refining and tipping out of it (the $K=3$ regressions visible in
 600 Section 4’s “Common failure modes” (4)), and when an unfair policy lands in the negative-maximin
 601 region, the alert injects exactly the rotation hint that recovers it.

602 D.4 Iteration logic ι : per-condition hyperparameters

603 Although ι has the smallest source surface, the researcher’s edits to it explain a meaningful fraction
 604 of the variance reduction noted in Finding 1. Table 6 summarizes the values shipped in the final
 605 commit of each best-performing run.

Table 6: Iteration-logic (ι) values in the final commit of selected runs. K = inner-loop iterations,
 $|S|$ = evaluation seeds, “thinking” = extended-thinking token budget passed to $\mathcal{M}$. Baseline ([3]) is
 $K=3$, $|S|=5$, thinking 16k.

Run	Game / objective	K	$ S $	thinking	Researcher’s stated rationale
exp1 (Gemini)	Cleanup, Φ_U	3	5	16k	default; stability comes from feedback hint
exp4 (Sonnet)	Cleanup, Φ_U	3	5	32k	“Sonnet was over-refining at 16k thinking”
exp5 (Gemini)	Cleanup, $\Phi_{\min}$	2	12	16k	“ $K=2$ avoids iter-3 regression; 12 seeds for variance control”
exp7 (Sonnet)	Cleanup, $\Phi_{\min}$	3	5	10k	“reduced thinking from 16k – Sonnet was generating runaway code at 16k”
gather-exp3 (Gemini)	Gathering, Φ_U	3	5	16k	default; Gathering converges in $K \leq 3$

606 The maximin Gemini run is the most informative case: the researcher discovered that with $K=3$
 607 and $|S|=5$, identical configurations could yield $\min_i R_i=295$ on one set of seeds and $\min_i R_i=100$
 608 on another (this is the “variance exposure” failure mode of Section 4’s common failures). It then
 609 walked $|S|$ from $5 \rightarrow 8 \rightarrow 12$ over three outer iterations until the seed-averaged signal was stable
 610 enough that the policy LLM stopped chasing noise. The policy in Listing 6 is sampled at $|S|=12$.

611 D.5 Cross-artifact observations

- 612 • *Helpers do the algebra; prompts pick the strategy class*. The researcher uses helpers (1, 2) to put
 613 non-trivial primitives at $\mathcal{M}$ ’s fingertips, and the prompt (3) to tell $\mathcal{M}$ *which* primitive to compose.
 614 Neither is sufficient alone — prompts without supporting helpers produced policies that “mention
 615 rotation” but failed to implement it; helpers without prompt updates often went unused.
- 616 • *Feedback is the only adaptive component*. p , $\mathcal{H}$, and ι are static within a run; ϕ is the only place
 617 where the researcher gets to change what $\mathcal{M}$ sees *during* the inner loop, and it uses thresholded
 618 diagnostics (Listing 4) to do so. This is what mostly drives the run-to-run variance reduction.
- 619 • *The same artifact has different content under each objective*. The Cleanup prompt under Φ_U omits
 620 “rotation” and “maximin” entirely; the same prompt under $\Phi_{\min}$ devotes its entire opening section

621 to it. The configuration c is genuinely a function of the welfare objective Φ , not a fixed pipeline
622 parameterized by it — consistent with the information-design framing of Section 3.4.
623 • *The researcher’s edits are short and targeted.* The full diff between the baseline pipeline and the
624 best maximin Gemini configuration totals ~ 300 added lines spread over four files; ~ 80 of them
625 appear in the listings above. The remaining content is API documentation and worked examples,
626 included for completeness but not strategically novel.

627 E Selected Generated Policies

628 The code excerpts below are verbatim outputs of the policy synthesizer $\mathcal{M}$, taken from the final
629 inner-loop iteration of the best-performing run in each condition. To fit the page, we elide repeated
630 boilerplate (BFS scaffolding, fallback branches) with `...` and add commentary in [brackets];
631 nothing else has been edited. The listings illustrate the qualitative differences highlighted in Findings
632 2–4 of Section 4: *static* vs. *rotating* cleaning roles in Cleanup, and the *Voronoi* + *respawn-timer*
633 structure in Gathering. All four policies were independently generated by the LLM under different
634 seeds, models, and outer-loop runs.

635 E.1 Cleanup, Φ_U (Gemini): static interleaved roles + dynamic threshold

636 Best run: $U=3.25$, $E=0.61$, $\min_i R_i = -182$. The synthesizer assigns *permanent* cleaning roles
637 via `agent_id mod  $n < \tau$` , where the threshold τ depends smoothly on the current waste fraction.
638 Higher pollution recruits more cleaners, but the *identity* of each cleaner is fixed — agents $0\text{--}\tau\text{--}1$
639 always pay the cost. This drives U to its ceiling but leaves equality at 0.61 and pushes the worst-off
640 agent to $\min_i R_i = -182$.

Listing 5: Cleanup, Φ_U – Gemini 3.1 Pro. Static role assignment with waste-tiered threshold (verbatim, abridged).

```

641 1 def policy(env, agent_id) -> int:
642 2     if int(env.agent_timeout[agent_id]) > 0:
643 3         return 7 # STAND while removed
644 4
645 5     waste_frac = waste_fraction(env)
646 6     n_agents = env.n_agents
647 7
648 8     # [Smooth dynamic threshold: more pollution -> more cleaners]
649 9     if waste_frac > 0.35: threshold_cleaners = n_agents
650 10    elif waste_frac > 0.25: threshold_cleaners = int(n_agents * 0.7)
651 11    elif waste_frac > 0.15: threshold_cleaners = int(n_agents * 0.5)
652 12    elif waste_frac > 0.05: threshold_cleaners = int(n_agents * 0.3)
653 13    else: threshold_cleaners = max(1, int(n_agents * 0.2))
654 14
655 15    # [STATIC role: cleaners are ALWAYS the lowest-id agents -- no rotation]
656 16    is_cleaner = (agent_id % n_agents) < threshold_cleaners
657 17
658 18    if is_cleaner:
659 19        best_orient, n_waste = find_best_clean_orientation(env, agent_id)
660 20        if n_waste >= 1:
661 21            cur_orient = int(env.agent_orient[agent_id])
662 22            if best_orient == cur_orient: return 8 # CLEAN
663 23            if (cur_orient + 1) % 4 == best_orient: return 5 # ROTATE_RIGHT
664 24            elif (cur_orient - 1) % 4 == best_orient: return 4 # ROTATE_LEFT
665 25            else: return 4
666 26
667 27            # [Competitor-aware BFS toward waste in own row band]
668 28            c_idx = agent_id % n_agents
669 29            target_r = int(((c_idx + 0.5) / threshold_cleaners) * env.height)
670 30            # ... pick waste cell minimizing dist + |row - target_r| + 50 *
671 31            # closer_cleaners
672 32            # ... bfs_toward and direction_to_action
673 33
674 34    else:
675 35        # GATHERER: row-banded sweep, competitor-aware
676 36        # assigned_row = ((g_idx + 0.5) / n_gatherers) * env.height
677 37        # ... pick alive apple minimizing dist + |row - assigned_row| + 50 *
678 38        # closer_gatherers
679 39        ...

```

680 E.2 Cleanup, $\Phi_{\min}$ (Gemini): time-rotated, geographically interleaved roles

681 Best run: $U=3.19$, $E=0.98$, $\min_i R_i=296$. Switching the outer objective from U to $\min_i R_i$ flips
682 the role-assignment scheme entirely: `role_idx` now depends on `agent_id + step/50`, so every
683 agent rotates through cleaner and gatherer duty in a 50-step cycle. The cleaner indices $\{1, 4, 8\}$ are
684 spatially *interleaved* along the river (rather than contiguous) so that whoever is on duty has a short
685 walk to their assigned river slice. This single change accounts for the $E=0.55 \rightarrow 0.98$ jump and the
686 $\min_i R_i=-182 \rightarrow 296$ swing reported in Table 2, with negligible efficiency loss.

Listing 6: Cleanup, $\Phi_{\min}$ – Gemini 3.1 Pro. Time-rotated roles with interleaved geography (verbatim, abridged).

```

687 1 def policy(env, agent_id) -> int:
688 2     if int(env.agent_timeout[agent_id]) > 0:
689 3         return 7 # STAND while removed
690 4
691 5
692 6     aid = int(agent_id)
693 7     step = int(env._step_count)
694 8     n = int(env.n_agents)
695 9
696 10    # [TIME ROTATION: every 50 steps each agent's role index advances]
697 11    if n == 10:
698 12        role_idx = (aid + step // 50) % 10
699 13        # [Cleaner indices {1,4,8} are SPATIALLY INTERLEAVED, not contiguous,
700 14        # so each on-duty cleaner has a short walk to its river slice]
701 15        if role_idx == 1: role_type, zone_idx = 'C', 0
702 16        elif role_idx == 4: role_type, zone_idx = 'C', 1
703 17        elif role_idx == 8: role_type, zone_idx = 'C', 2
704 18        else:
705 19            role_type = 'G'
706 20            g_map = {0:0, 2:1, 3:2, 5:3, 6:4, 7:5, 9:6}
707 21            zone_idx = g_map[role_idx]
708 22            num_cleaners, num_gatherers = 3, 7
709 23
710 24    if role_type == 'C':
711 25        # [Each on-duty cleaner owns one horizontal slice of the river]
712 26        row_min = int( zone_idx * env.height / num_cleaners)
713 27        row_max = int((zone_idx + 1) * env.height / num_cleaners) - 1
714 28
715 29        # [Score every (row, col, orientation) in the slice by beam yield;
716 30        # fire if facing >=2 waste, else move to a >=2-yield position]
717 31        best_positions, good_positions = set(), set()
718 32        for r in range(row_min, row_max + 1):
719 33            for c in range(10):
720 34                if env.walls[r, c] or env.waste[r, c]: continue
721 35                y_max = max(get_clean_yield(env, r, c, o) for o in range(4))
722 36                if y_max >= 2: best_positions.add((r, c))
723 37                if y_max >= 1: good_positions.add((r, c))
724 38            # ... rotate / CLEAN / bfs_to_target_set as appropriate
725 39        else:
726 40            # GATHERER: collect apples ONLY inside the rotating row band
727 41            # row_min/row_max for num_gatherers slices ...
728 42            ...

```

730 E.3 Cleanup, $\Phi_{\min}$ (Sonnet): independent rediscovery of duty rotation

731 Best run: $U=2.93$, $E=0.83$, $\min_i R_i=154$. Run with a different policy LLM (Sonnet 4.6) and on
732 a different outer-loop trajectory, the synthesizer arrives at the *same* structural insight as the Gemini
733 maximin run: a phase counter (`agent_id + step/50`) mod n rotates which 2 of the 10 agents clean
734 at any given time, and a *separate* 200-step zone counter rotates which 5-row band each collector
735 sweeps. The two rotation periods (50 and 200 steps) ensure every agent visits every cleaning slot
736 *and* every apple zone within one episode — a structural fairness invariant the researcher discovered
737 without any explicit instruction to “rotate.”

Listing 7: Cleanup, $\Phi_{\min}$ – Sonnet 4.6. Phase-rotated cleaning + zone-rotated collection (verbatim, abridged).

```

738 1 def policy(env, agent_id) -> int:
739 2     if int(env.agent_timeout[agent_id]) > 0:
740 3         return 7 # STAND while removed
741 4
742 5
743 6     n = env.n_agents
744 7     step = env._step_count
745 8     wf = waste_fraction(env)

```

```

746 8
747 9 # --- ROLE ASSIGNMENT: fair 2-cleaner rotation ---
748 10 # [50-step phases x 20 phases / episode -> each agent cleans 4 phases (~20%)]
749 11 phase = step // 50
750 12 cleaner_rank = (agent_id + phase) % n # [cycles 0..9 fairly]
751 13 is_cleaner = cleaner_rank < 2 # [exactly 2 cleaners per phase]
752 14
753 15 # [Emergency override: all agents clean when waste crosses the spawn cliff]
754 16 in_emergency = wf >= 0.40
755 17 if in_emergency: is_cleaner = True
756 18
757 19 # [River split: cleaner_slot 0 -> top half, 1 -> bottom half]
758 20 cleaner_slot = (agent_id % 2) if in_emergency else cleaner_rank
759 21
760 22 if is_cleaner and wf > 0.04:
761 23     # [Scan 4 orientations from current position; if best beam shot
762 24     # covers >=3 waste cells, fire; else step to a strictly better neighbor]
763 25     # ... (best_o, best_cnt) = max over 4 directions
764 26     # ... if best_cnt >= 3: rotate to best_o, then CLEAN
765 27     # ... else: scan 4 adjacent cells x 4 orientations for higher yield
766 28     # ... if no waste in beam range: bfs to waste_set in OWN river half,
767 29     # fall back to whole river
768 30
769 31 # --- COLLECTOR LOGIC: 5-zone rotating apple collection ---
770 32 # [5 zones x 200 steps = 1000 steps -> each agent visits every zone once;
771 33 # 2 agents share each zone at any time, reducing competition 10 -> 2]
772 34 bounds = zone_boundaries(env, 5)
773 35 zone_phase = step // 200
774 36 zone = (agent_id + zone_phase) % 5
775 37 z_start, z_end = bounds[zone], bounds[zone + 1]
776 38
777 39 zone_apples = {(int(env._apple_pos[i][0]), int(env._apple_pos[i][1]))
778 40                for i in range(env.n_apples) if env.apple_alive[i]
779 41                and z_start <= int(env._apple_pos[i][0]) < z_end}
780 42 # ... bfs_to_target_set(zone_apples), else bfs_nearest_apple as fallback
781 43 return 7

```

783 E.4 Gathering (Gemini): wall-aware Voronoi + spatiotemporal targeting

784 Best run: $U=2.47$, $E=0.98$, $\min_i R_i=580$. In Gathering, where costs are symmetric, no role dif-
785 ferentiation is needed: the entire optimization is spatial and temporal. The synthesizer (i) parti-
786 tions cells into BFS-Voronoi territories owned by individual agents, (ii) runs a single centralized
787 BFS from its own position to score every reachable cell with exact wall-aware distances, and
788 (iii) for both alive and respawning apples in its territory, computes the *earliest collection time*
789 $\max(\text{walk_dist}, \text{respawn_timer})$ and targets the minimum. When the agent must wait on a
790 dead apple, it steps onto a *non-spawn* adjacent cell rather than blocking a respawn point. This single
791 sort key — $\max(\text{distance}, \text{timer})$ — subsumes both “go to nearest apple” and “camp respawn” as
792 special cases.

Listing 8: Gathering – Gemini 3.1 Pro. Voronoi territories + spatiotemporal priority targeting (ver-
batim, abridged).

```

793 1 def policy(env, agent_id) -> int:
794 2     if int(env.agent_timeout[agent_id]) > 0:
795 3         return 7
796 4
797 5 r, c = int(env.agent_pos[agent_id][0]), int(env.agent_pos[agent_id][1])
798 6 orient = int(env.agent_orient[agent_id])
799 7
800 8 # [Wall-aware Voronoi over the gridworld; spawn_points = all apple cells]
801 9 zones = voronoi_zones(env)
802 10 spawn_points = {(int(p[0]), int(p[1])) for p in env._apple_pos}
803 11
804 12 # [Single centralized BFS: exact distance + first move to every reachable cell]
805 13 distances, first_moves = {}, {}
806 14 q = deque([(r, c, 0, None)])
807 15 visited = {(r, c)}
808 16 while q:
809 17     cr, cc, d, fm = q.popleft()
810 18     distances[(cr, cc)] = d
811 19     first_moves[(cr, cc)] = fm
812 20     for dr, dc in [(-1,0), (1,0), (0,-1), (0,1)]:
813 21         nr, nc = cr + dr, cc + dc
814 22         if 0 <= nr < env.height and 0 <= nc < env.width and not env.walls[nr,
815 23             nc]:
816 24             if (nr, nc) not in visited:

```

```

818 24         visited.add((nr, nc))
819 25         q.append((nr, nc, d + 1, fm if fm is not None else (dr, dc)))
820 26
821 27     # [Both alive AND respawning apples in MY zone, with their respawn timer]
822 28     my_zone_spawns = []
823 29     for i in range(env.n_apples):
824 30         pr, pc = int(env._apple_pos[i][0]), int(env._apple_pos[i][1])
825 31         if zones.get((pr, pc)) == agent_id and (pr, pc) in distances:
826 32             my_zone_spawns.append((pr, pc, int(env.apple_timer[i]), distances[(pr,
827 33                                     pc)]))
828 34
829 35     # [SPATIOTEMPORAL PRIORITY: earliest collection time first.
830 36     # score = max(walk_distance, respawn_timer); ties broken by sooner timer]
831 37     my_zone_spawns.sort(key=lambda x: (max(x[3], x[2]), x[2], x[3]))
832 38
833 39     for pr, pc, timer, dist in my_zone_spawns:
834 40         if timer == 0: # [Alive apple: walk straight to it]
835 41             if dist == 0: return 7
836 42             fm = first_moves[(pr, pc)]
837 43             return direction_to_action(fm[0], fm[1], orient)
838 44         else: # [Dead apple: camp on safe adjacent cell]
839 45             # ... pick nearest neighbor (nr,nc) that is NOT in spawn_points,
840 46             # navigate there and STAND while waiting for respawn
841 47             ...
842 48     # [Global poaching fallback if zone is empty: nearest alive apple anywhere]
843 49     ...

```

845 E.5 Cross-condition observations

846 Several patterns are visible across these four listings (and confirmed by inspecting the remaining 8
847 runs not shown):

- 848 • *Role assignment encodes the welfare objective.* Static $\text{agent_id} < \tau$ (Listing 5) maximizes
849 U but harms $\min_i R_i$. Time-rotated $(\text{agent_id} + \text{step} // T) \% n$ (Listings 6, 7) maximizes
850 $\min_i R_i$ at $\leq 1\%$ efficiency cost (Gemini).
- 851 • *Convergent rediscovery across LLMs.* Gemini and Sonnet, on independent runs, both arrive at
852 the $(\text{id} + \text{phase}) \% n$ duty-rotation idiom with similar phase lengths ($T \approx 50$ steps) under
853 maximin — and both omit it under efficiency.
- 854 • *Game structure dictates strategy class.* Cleanup policies are dominated by role-assignment logic
855 (cleaner vs. gatherer); Gathering policies are dominated by territory partitioning and respawn-
856 timer reasoning, with no role differentiation at all. The researcher does not need to be told which
857 class of strategy to write.
- 858 • *Earliest-collection-time as a unifying score.* The Gathering policy’s $\max(\text{walk}, \text{timer})$ key (List-
859 ing 8) is structurally identical across the 4 Gathering runs (both Gemini and Sonnet), differing
860 only in how the Voronoi diagram is computed — Manhattan approximation, fully BFS-based, or
861 cached helper.